# FYS4480/9480 October 22 and 23, 2026

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$N = 2$

$$\int dx_1 dx_2 \; \psi^*(x_1, x_2) \frac{1}{|x_1 - x_2|} \psi(x_1, x_2)$$

$$< \sum_{i<j} v(x_i, x_j) >$$

$N = 3$

$$\int dx_1 dx_2 dx_3 \; \psi^*(x_1, x_2 x_3) \frac{1}{(x_1 - x_2|} \times \psi(x_1, x_2 x_3)$$

$$+ \int dx_1 dx_2 dx_3 \; \psi^*(x_1 x_2 x_3) \frac{1}{|x_1 - x_3|} \times \psi(x_1, x_2 x_3)$$

$$+ \int dx_1\, dx_2\, dx_3\; \psi^*(x_1 x_2 x_3)\, \frac{1}{|x_2 - x_3|}\, \psi(x_1 x_2 x_3)$$

Define $\rho_2(x_1, x_2) = \int dx_3\, |\psi|^2$

$$\rho_2(x_1, x_3) = \int dx_3\, |\psi|^2$$

$$\rho_2(x_2 x_3) = \int dx_1\, |\psi|^2$$

$$\langle V \rangle = \int dx_1\, dx_2\, \frac{\rho_2(x_1, x_2)}{|x_1 - x_2|}$$

$$+ \int dx_1\, dx_3\, \frac{\rho_2(x_1, x_3)}{|x_1 - x_3|} + \int dx_2\, dx_3\, \frac{\rho_2(x_2 x_3)}{|x_2 - x_3|}$$

$3! = 1 \cdot 2 \cdot 3 = 6$

in general

$$\langle U \rangle = \frac{N(N-1)}{2} \int dx\, dx' \times \frac{\rho_2(x, x')}{|x - x'|}$$

$$\langle V_{ext} \rangle = \int dx\, n(x)\, v_{ext}(x)$$

$$= V_{ext}[n]$$

$n(x)$: one-body density

$$\langle \hat{H} \rangle = E = T[n] + v_{ext}[n] + v[?]$$

$$\langle \hat{H} \rangle = E[\Psi]$$

FCI
HF
MBPT
CC / Green's fun ctn

$\longrightarrow$ wave function based methods

$$\langle \hat{H} \rangle = E[n]$$

DFT

↳ Hohenberg-Kohn

↳ Kohn-Sham eqs.